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How to repeat a character 'n' times in Bash

Author: Vivek Gite • Last updated: October 18, 2021 • 10 comments

Let us say you want to repeat a character such as '-' OR '=' multiple times while writing bash script. Please note that I needed a quick script to work on Linux, macOS, and FreeBSD server to repeat a string/character n times. One simple way to to repeat a string or character n times is simple use the [echo command/printf command](#) as follows:

```
echo '-----'
```

OR

```
printf '%s\n' '-----'
```

Tutorial details	
Difficulty level	Easy
Root privileges	No
Requirements	Bash/Ksh on Linux/Unix
Est. reading time	2 minutes

But, there must be a better way. For instance, here are classic Perl and Python examples that repeat '-' 80 times:

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```
perl -E 'say "-" x 80'
```

Python one-liner:

```
python -c 'print "-" * 80'
```

How to repeat a character 'n' times in Bash

Let us try this example Linux and Unix Bash function to repeat a string/character 'n' times with help of [bash for loop](#) as follows:

```
# Repeat given char 80 times using shell function
repeat(){
    for i in {1..90}; do echo -n "$1"; done
}
```

Then all I have to do is call repeat() as follows:



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```
repeat '-'
echo

repeat '='
echo

repeat '~'; echo -e '\nHEAD 1\t\tHead 2\t\tHead 3\t\t\t\t\tHead 4'; repeat '~'; echo
```

```
vivek@nixcraft-wks01 tmp $ # Repeat given char 80 times using shell function
vivek@nixcraft-wks01 tmp $ repeat(){
> for i in {1..90}; do echo -n "51"; done
> }
vivek@nixcraft-wks01 tmp $ repeat '-'; echo
vivek@nixcraft-wks01 tmp $ repeat '='; echo
vivek@nixcraft-wks01 tmp $ repeat '~'; echo -e '\nHEAD 1\t\tHead 2\t\tHead 3\t\t\t\t\tHead 4'; repeat '~'; echo
HEAD 1      Head 2      Head 3      Head 4
vivek@nixcraft-wks01 tmp $
vivek@nixcraft-wks01 tmp $ echo $BASH_VERSION
5.0.17(2) release
vivek@nixcraft-wks01 tmp $
```

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My solution is simple and friendly. It solved my problem. Yet now I am getting greedy. I wanted to see if I could use printf/echo and replicate the Perl/Python one-liner experience.



Repeating a character or string n times using printf

The `printf` is a shell built-in and external `/bin/printf`, used to format and prints ARGUMENTS. The syntax is:

```
printf 'FORMAT' var
printf '%s\n' 'Hello World'
printf -- '-%.0s' 1 2 3 4 5
printf -- '-%.0s' {1..5}
printf -- '-%.0s' {1..80}
```

Where,

- The **double dash** “`--`” means “end of command line flags.” Since I need to print `-`, the double dash (`--`) tells printf command not to try to parse what comes after command line options.
- Bash expands `{1..5}`. For example, `echo {1..5}`, prints `1 2 3 4 5`. So `printf -- '-%.0s' {1..5}` becomes: `printf -- '-%.0s' 1 2 3 4 5`
- The `%.0s` print single ‘`-`’ given 5 times.
- You can remove double dash (`--`) when not printing ‘`-`’. For instance: `printf '~%.0s' {1..80}; printf '\n'`

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Closing note

I ended up using the following as I needed to run script on both macOS/BSD and Linux:

```
# Repeat given char N times using shell function
repeat(){
  local start=1
  local end=${1:-80}
  local str="${2:-=}"
  local range=$(seq $start $end)
  for i in $range ; do echo -n "${str}"; done
}
```

Then I run it as follows:

```
repeat 80 '-'; echo

repeat 40 '='; echo

# print Hello 5 times
repeat 5 'Hello'; echo
```

I am sure someone could come up with awk, sed, echo/printf and other versions too. But, for now, I am satisfied with what I got. As usual, check man pages using the [help command](#) or [man command](#):

```
man bash
help printf
man seq
```

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Harshvardhan Gupta • Jun 4, 2021 @ 11:25

You can use powershell core on Linux and write: “*” * 25

[link](#)

Vivek Gite • Jun 5, 2021 @ 20:26

No.

[link](#)

Bit Twiddler • Jun 5, 2021 @ 8:47

Nice! I'd previously used things like:

```
head -c 80 < /dev/zero | tr '\0' '.'
```

[link](#)

Anonymous • Jun 6, 2021 @ 19:48

Wow. Never thought of that.

[link](#)

Cüneyt Egin • Jun 5, 2021 @ 18:44

How about:

```
seq -s_ 200|tr -d '[:digit:]'
```

to print a line with 200 underscores...

[link](#)

Dumitru Uzun • Jun 6, 2021 @ 5:40

There is a small typo in the last line of the last 'repeat()':
It should be "\$str" instead of "\$2"

[link](#)

Vivek Gite • Jun 6, 2021 @ 8:30

Fixed it. Thank you kind stranger.

[link](#)

Rehman • Jun 6, 2021 @ 12:25

As usual, awesome post for Bash lovers like me.

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Didi • Jun 6, 2021 @ 19:48

+1 from me too.

[link](#)

Bruce M • Jun 9, 2021 @ 11:00

```
#!/bin/sh
# repeat times char
if test $# -ne 2 ; then
    echo 1>&2 usage: $(basename $0) times char
    exit 1
fi
yes "$2" | head -n "$1" | tr -d '\n'
```

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